

# AMRITANSHU BARPANDA

DOB: 30 January, 2005      Rourkela, Sundergarh, Odisha       Portfolio Website  
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## Education

### National Institute of Technology, Rourkela

*Bachelor of Technology in Electronics and Instrumentation Engineering (CGPA - 7.76)*

**August 2023 – Present**

*Rourkela, Odisha*

### Delhi Public School, Sambalpur

*AISSE – CBSE, Science (PCM) (Percentage - 90%)*

**May 2023**

*Sambalpur, Odisha*

### Delhi Public School, Rourkela

*AISSE – CBSE (Percentage - 96%)*

**May 2021**

*Rourkela, Odisha*

## Relevant Coursework

- Data Structures and Algorithms
- Object Oriented Programming
- Database Management Systems

## Work Experience

### Rourkela Steel Plant

**May 2025 – June 2025**

*Vacational Trainee*

*Certificate*

- Designed and implemented a battery-powered ESP32-S3-based SoftAP module that monitors and transmits temperature and humidity data over Wi-Fi and Ethernet for Electrical Control Rooms at RSP, ensuring real-time environmental monitoring during power outages.

### GRID-INDIA

**June 2024 – July 2024**

*Summer Intern*

*Certificate*

- Developed a program to generate detailed map infographics by processing power plant coordinates, cost per unit, and capacity, enabling clear visualization of energy distribution.
- Conducted performance analysis of Quadratic vs Linear Programming in **Price-Based Economic Dispatch (PBED)** systems; found QP slightly more efficient, but not significantly better to warrant replacing existing LP models.

## Projects

### Parampara

**December 2025**

*React, Node.js, Express, Supabase, REST APIs*

*Website*

- Developed a full-stack community-driven web platform enabling users to document, share, and explore family rituals across regions, featuring secure authentication, user profiles, ritual creation and discovery.

### Om tours

**March 2025**

*Python, HuggingFace, Smolagents, APIs*

*Website*

- Developed a full-stack agentic AI system using Smolagents, FastAPI, and React to create personalized travel itineraries through multistep decision-making, covering budget, transport, and schedule, with an interactive frontend for visualization.

### Next-Problem

**September 2024 – December 2024**

*Python, React Js, Vercel, API*

*Website — GitHub*

- Designed and deployed **Next-Problem**, a personalized coding question recommendation tool leveraging **Elo-MMR** to filter and match problems to user skill levels. Engaged **250+** users and recorded **500+** visits in the first month.

## Achievements

### • Hacktoberfest

**Oct 2023**

Contributed 6 pull requests, 4 of which were accepted and merged into open-source repositories, showcasing collaboration skills.

Treenation — Badges — Certificate

## Technical Skills

- **Languages:** Python, HTML, CSS, Javascript, SQL(Structured Query Language), C++
- **Developer Tools:** VS Code, Jupyter Notebook, Google Colab, Spyder, Postman
- **Libraries:** Numpy, Pandas, Matplotlib, Pytorch, Web Scraping (BeautifulSoup/Selenium)

## Certifications

### Artificial Intelligence and Robotics

**June 2025 – July 2025**

*IIIT, Allahabad*

*Certificate*

- Underwent a 6-week training program on AI and Robotics at IIIT Allahabad, gaining practical exposure to machine learning, computer vision, and autonomous robotic systems.